META-ANALYSIS OF TEMPORAL GENE EXPRESSION PROFILES IN FETAL HEART DEVELOPMENT

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AIM

To identify robust differential expression patterns of genes during fetal heart development.

METHODOLOGY

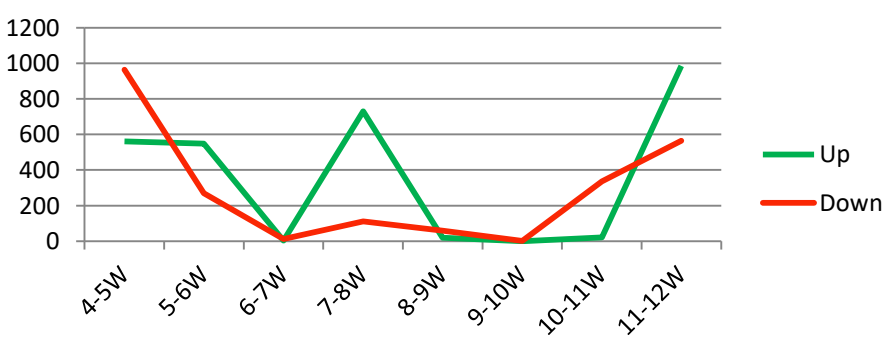
1. Analysis of RNAseq expression data
2. Identification of Differentially expressed genes
3. Differential expression and significant gene identification over each week of development
4. Identification and detailed study of expression pattern of differentially expressed genes in CHDs
5. Elucidate the importance of SNPSs in CHDs using bioinformatics tools
6. Ontology and enrichment analysis
7. Pathway Analysis

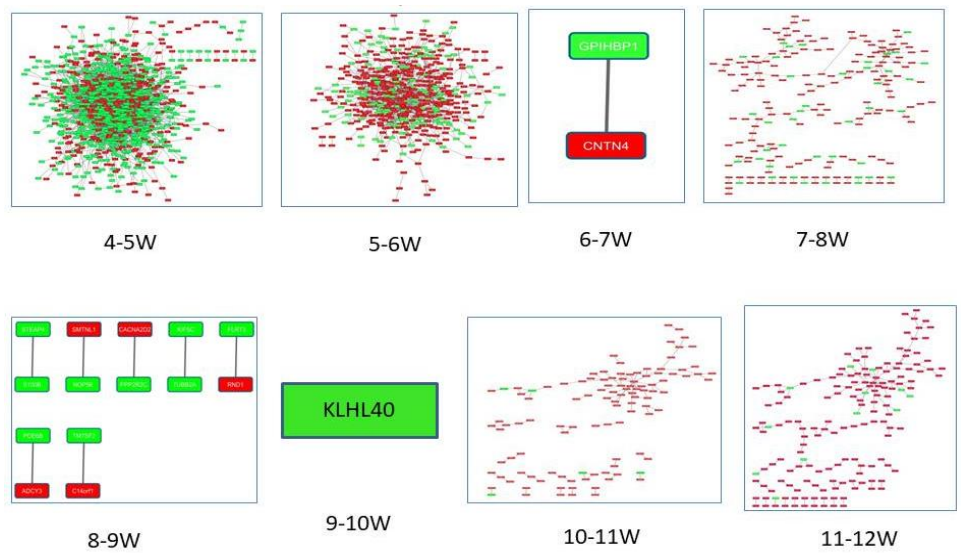
DATASETS

Week 4	Week 5	Week 6	Week 7	Week 8
• ERS2510274 • ERS2510273	• ERS2510288 • ERS2510289	• ERS2510300 • ERS2510299	• ERS2510317 • ERS2510316 • ERS2510315	• ERS2510340 • ERS2510339 • ERS2510338 • ERS2510336

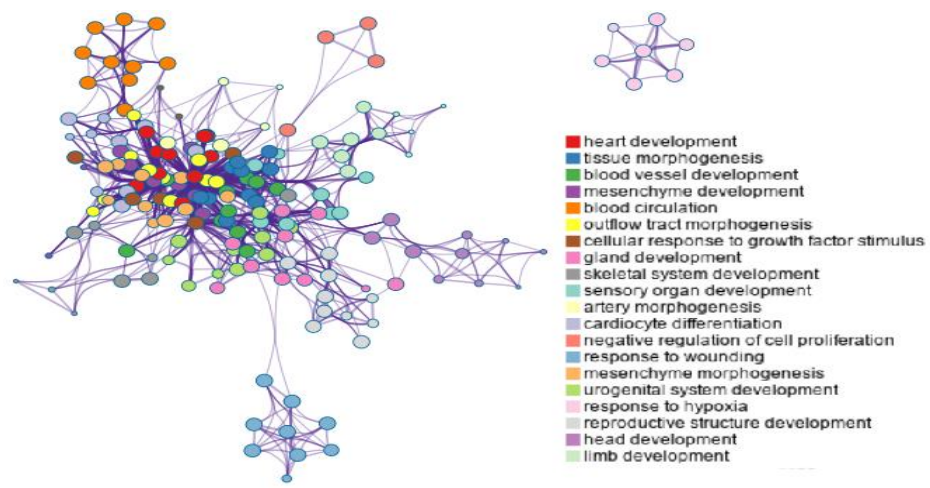
Week 9	Week 10	Week 11	Week 12
• ERS2510361 • ERR2704718 • ERR2704719 • ERR2704720	• ERS2510161 • ERS2510160 • ERS2510158 • ERS2510159	• ERS2510179 • ERS2510182 • ERS2510181 • ERS2510180	• ERS2510197 • ERR2704712 • ERR2704713 • ERR2704714

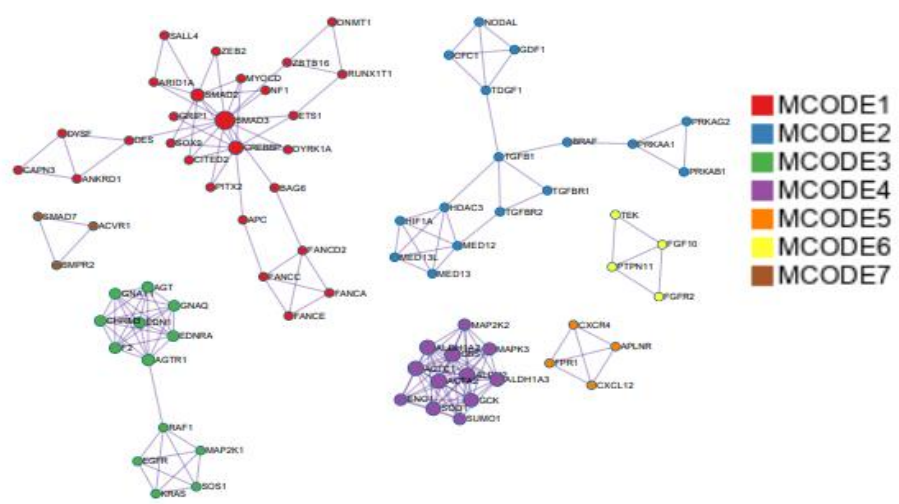
RESULTS





Week by week differential expression analysis





CONCLUSION

- 326 DEGs and 26 lncRNAs were found to be involved in Congenital Heart Disorders.
- 92 linked SNPs were identified which had association with other cardiovascular diseases.